

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
II SEMESTER 2025-26
CS/ECE/EEE/INSTR F241 MICROPROCESSOR PROGRAMMING
17-04-2026 (Friday) QUIZ-5(closed book)

A

Max time: 10 Min
ID No: _____

Name: _____

Max Marks: 08
Sec No.: _____

1. In an 8086 based system, when an INTR interrupt is raised (when INTR is acknowledged, the interrupting hardware places type number FOH on the data bus.), the LED connected to Port C.4 of 8255 should be turned on (LED SHOULD glow) via BSR Mode. The LED is connected in common anode configuration. The base address of 8255 is FFAFh (Use FFAFh and next possible addresses in increasing sequence for 8255). Answer the following question

- a. From which address (s) will the IP of the ISR fetched [1M]

0000: 03C0 or 003C0 — 0.5
 0000: 03C1 003C1 — 0.5

- b. What is the address of Port B? FFBFH [1M]

- c. Mention the lines of code (use minimum lines) required to initialize Port C [2M]

MOV AL, 1xxxxxxx — 1M
 MOV DX, FFFFH
 Out DX, AL] 1M

- d. Mention the lines of code (use minimum lines) required to turn on LED connected to PC.4 on an interrupt [2M]

MOV AL, 0xxx1000 — 1M
 MOV DX, FFFFH
 Out DX, AL] 1M

- e. Which address line from 8086 will be connected to the following A1 pin of 8255
A6 [1M]

- f. Which data lines (mention range (starting line-ending line)- eg D17-D24) from 8086 will be connected to the data lines of 8255 D8-D15 [1M]

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
1-I/O Mode	Port A Mode 00-Mode0 01-Mode1 1x-Mode2	Port A 1-I/p 0-o/p	Port C Upper 1-I/p 0-o/p	Port B Mode 0-Mode0 1-Mode1	Port B 1-I/p 0-o/p	Port C Lower 1-I/p 0-o/p	
	Group A			Group B			

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀						
0-BSR	x	x	x	Bit ₂	Bit ₁	Bit ₀	Bit Set/Reset						
Don't Care Condition				PC	0	1	2	3	4	5	6	7	1 - Set 0 - Reset
				B ₀	0	1	0	1	0	1	0	1	
				B ₁	0	0	1	1	0	0	1	1	
				B ₂	0	0	0	0	1	1	1	1	

A1
 A 1010
 B 1011
 C 1110
 D 1111